

# Resolution Geometry Made Visible

## The Single Origin and Its Stratified Space

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July 18, 2026

### Abstract

Resolution geometry (RG) reduces, object by object, to established mathematics, yet no smooth manifold houses it. This paper resolves that tension by drawing the object. The resolution is not that RG lacks a home in mathematics but that its home is one category above manifolds: a *Whitney-stratified orbit-space bundle*. Everything issues from a single origin—one Schur elimination on one object  $\mathcal{G} = (\mathcal{O}, g; J, \Sigma)$ —which splits into three sectors carrying three different geometries: the resolved sector is Riemannian (Fisher distinguishability, genuine paths); the coupling is angular (canonical correlations are cosines of principal angles between subspaces, a stratified Grassmannian); the null sector is conical (the Schur complement and second fundamental form live in the cone of positive-semidefinite forms, stratified by rank). Assembled up to gauge, a single resolution geometry is a point in a product of closed eigenvalue chambers cut by the Schur admissibility inequality—an orbit space with walls where eigenvalues collide, conical singularities at the degeneracies (carrying the  $\mathbb{Z}/2$  diabolical phase), and a boundary at the admissibility floor. Over the smooth parameter base these stratified fibres assemble into a bundle: paths, transport, holonomy, and monodromy live along the base; eigenvalues, angles, and cones live in the fibre. Why one could never find the manifold is now visible—the honest object is stratified, its non-smoothness is its content, and every piece (orbit spaces, Grassmannians, PSD cones, Luna slices) is standard. This is what makes RG a geometry in the full modern sense. All figures are drawn; no claim is physical.

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# 1 The question: no manifold, and why that is the answer

Two facts sit in apparent contradiction. First, every object of RG reduces to established mathematics; nothing is a new primitive. Second, no smooth manifold carries the whole structure: the resolved sector wants a metric, the null sector refuses one, and the strata where the two meet are singular. The temptation is to read the second fact as a gap in the first—as if RG failed to be locatable in mathematics.

It is the opposite. Established mathematics is far larger than the category of manifolds. Orbit spaces, convex cones, Grassmannians with their Schubert strata, and Whitney-stratified sets are all rigorous and all standard, and none is a smooth manifold. “No manifold works” is therefore not a failure to place RG; it is the discovery that the correct object is *stratified*. The non-smoothness one keeps bumping into is not an obstruction to drawing the geometry—it is the geometry. This paper draws it.

## 2 The single origin: one object, one elimination

Everything below issues from one object and one operation. Fix an operating point with observation map  $J$  into an observation space  $(\mathcal{O}, g)$  and covariance  $\Sigma$ . The resolved sector is  $R = \text{Im } J$ ; the null sector is  $N = R^\perp$ ; and  $\Sigma$  splits against  $\mathcal{O} = R \oplus N$  into blocks. A single Schur elimination on that block matrix produces all three sectors and their geometries at once.

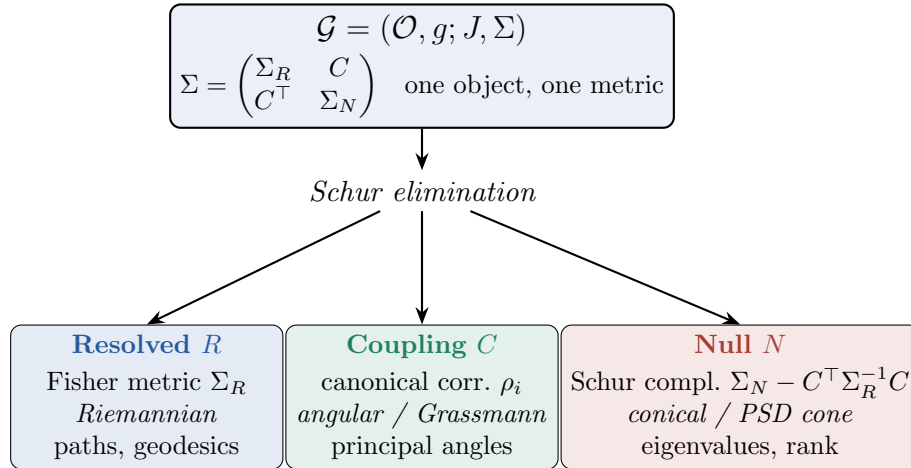


Figure 1: The single origin. One object  $\mathcal{G}$  and one Schur elimination produce three sectors, each carrying a *different* kind of geometry. No single smooth structure covers all three; that is why RG is stratified, not a manifold.

The rest of the paper reads off Figure 1: §3 draws each sector’s geometry, §4 assembles them into the fibre orbit space, and §5 bundles the fibre over the base.

## 3 Three sectors, three geometries

### 3.1 Resolved $\rightarrow$ Riemannian: paths of distinguishability

The resolved sector inherits the Fisher metric  $g_F = J^\top \Sigma^{-1} J$ . A path here is a curve of *distinguishability*: its length measures how many samples separate two nearby models. This is genuine Riemannian geometry—smooth, with geodesics—and it is the only sector that has paths, because it is the only sector in which motion changes what is observed.

### 3.2 Coupling → angular: principal angles between subspaces

The coupling does not carry paths; it carries *angles*. The canonical correlations  $\rho_i$  are the cosines of the principal angles between the resolved subspace and the informative directions of the covariance. Their home is a Grassmannian—the space of subspaces—which is stratified by rank, not smooth throughout.

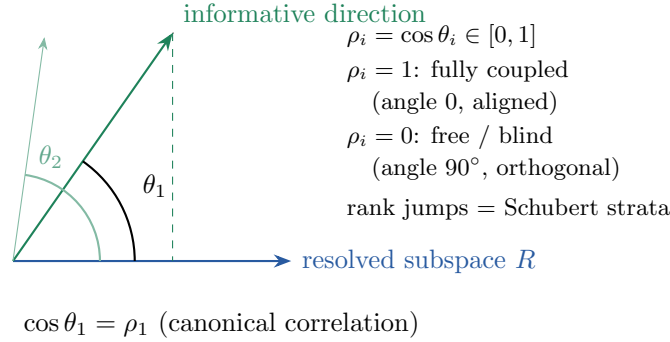


Figure 2: The coupling is angular. Each canonical correlation is the cosine of a principal angle between the resolved subspace and an informative direction. The invariant is an angle, not a path; the space of such configurations is a stratified Grassmannian.

### 3.3 Null → conical: the cone of positive-semidefinite forms

The null sector carries neither paths nor angles but *quadratic forms*: the residual covariance (the Schur complement) and the second fundamental form are positive-semidefinite, and their home is the PSD cone. A cone is convex but singular: its interior is the full-rank forms, its curved boundary is the rank-deficient ones, and its apex is the zero form. Rank is the stratification.

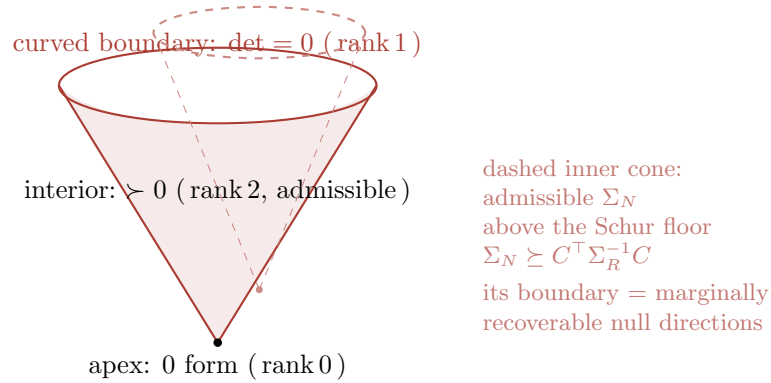


Figure 3: The null sector is conical. Residual covariance and second fundamental form live in the PSD cone, stratified by rank: interior (full rank), curved boundary ( $\det = 0$ ), apex (zero). The Schur floor is a shifted sub-cone (dashed); its boundary is where null directions become marginally recoverable. The invariant is an eigenvalue/rank, not a path.

Three sectors, three geometries: metric, angular, conical. No single smooth chart carries a metric, an angle stratification, and a cone at once; the object that does is stratified.

## 4 The fibre: the orbit space of one resolution geometry

Assemble the three sectors and quotient by gauge (independent orthogonal change in the resolved and null frames). Up to gauge, a single resolution geometry is described by a short,

honest coordinate list.

**Proposition 4.1** (Fibre coordinates). *A resolution geometry, modulo the gauge group  $O(p) \times O(q)$ , is determined by*

$$\left( \underbrace{\lambda_1 \geq \cdots \geq \lambda_p \geq 0}_{\text{resolved spectrum}}, \underbrace{\mu_1 \geq \cdots \geq \mu_q \geq 0}_{\text{null spectrum}}, \underbrace{\rho_1 \geq \cdots \geq \rho_k \geq 0}_{\text{canonical correlations}} \right), \quad \Sigma_N \succeq C^\top \Sigma_R^{-1} C,$$

*a point in a product of closed eigenvalue chambers cut by the Schur admissibility inequality.*

Each spectrum lives in a closed Weyl chamber  $\{\lambda_1 \geq \cdots \geq \lambda_n \geq 0\}$ . A chamber is not a smooth manifold: its *walls*, where consecutive eigenvalues coincide, are singular strata.

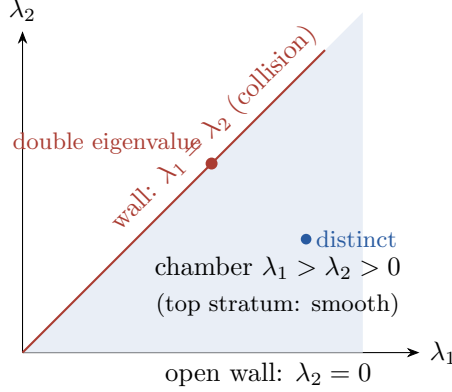


Figure 4: Each spectrum lives in a closed Weyl chamber (shaded). The interior is the smooth top stratum (distinct eigenvalues); the diagonal wall  $\lambda_1 = \lambda_2$  is a singular stratum where eigenvalues collide. The fibre is a product of such chambers, cut by the Schur floor.

At the walls the geometry develops *conical singularities*, and these are not cosmetic: a loop of the null frame around a double-eigenvalue point returns with a sign, the  $\mathbb{Z}/2$  diabolical phase (verified elsewhere in the corpus as a holonomy of exactly  $-1$ ). The double cone is the local picture.

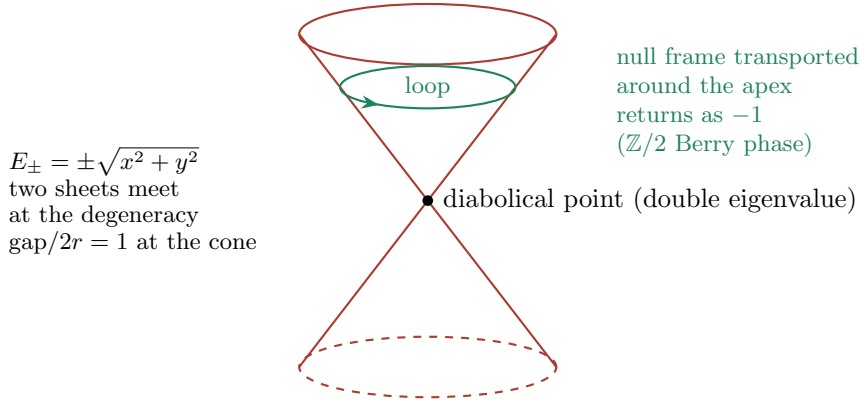


Figure 5: The local model at a wall: a double cone. The two eigenvalue sheets meet at the diabolical point (the degeneracy); a loop of the null frame around the apex returns multiplied by  $-1$ . Singularities carry invariants—this one is a genuine  $\mathbb{Z}/2$  holonomy, not an artifact.

The fibre is therefore a *stratified orbit space*: a product of closed chambers, cut by the admissibility cone, with walls (eigenvalue collisions), conical singularities (degeneracies, carrying the diabolical phase), and an outer boundary (the Schur floor). Its local models at each stratum are Luna slices; its natural cohomology is intersection cohomology, the theory built for exactly such spaces.

## 5 The final object: a stratified bundle over a smooth base

One fibre is a stratified orbit space. The full object is the family of these fibres over the smooth parameter base, glued by transport. This is where the two kinds of geometry separate cleanly, and it answers the original worry about paths.

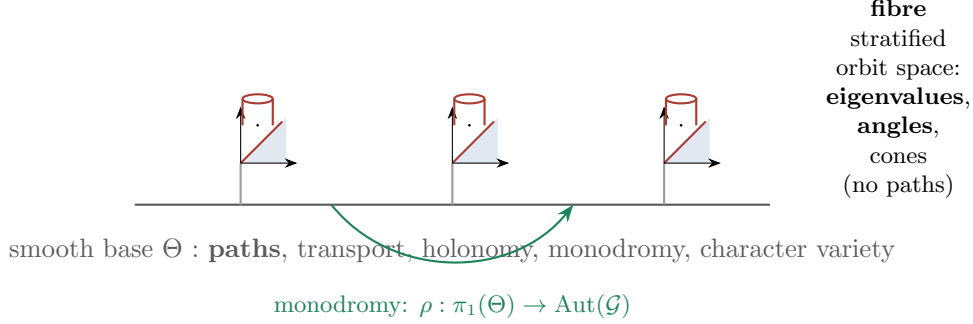


Figure 6: The final object: a stratified orbit-space bundle. The base is smooth and carries all the *paths*—transport, holonomy, monodromy, the character variety. Each fibre is the stratified orbit space of §4 and carries *eigenvalues*, *principal angles*, and *cones*—no paths. Fisher-style paths and the blind sector never competed; they are the base and fibre directions of one object.

**Proposition 5.1** (Where paths live, where eigenvalues live). *In the stratified bundle of Figure 6, path-type invariants (geodesics, holonomy, monodromy) are base-direction phenomena on the smooth top stratum, while spectral-type invariants (eigenvalues, principal angles, rank, cone position) are fibre-direction phenomena on a stratified space that admits no paths in its singular directions. The two are the horizontal and vertical geometries of one bundle.*

This is why the search for a single manifold failed. A manifold would force one smooth geometry everywhere; the honest object has smooth path-geometry along the base and stratified spectral geometry in the fibre, meeting at singular walls. The following table is the whole picture in one view.

|                                  | invariant type                            | home                    |
|----------------------------------|-------------------------------------------|-------------------------|
| resolved sector (base)           | paths, geodesics, distinguishability      | Riemannian (Fisher)     |
| transport / globalization (base) | holonomy, monodromy                       | character variety       |
| coupling (fibre)                 | principal angles $\rho_i = \cos \theta_i$ | stratified Grassmannian |
| null sector (fibre)              | eigenvalues, rank                         | PSD cone                |
| degeneracy walls (fibre)         | $\mathbb{Z}/2$ diabolical phase           | conical singularity     |
| admissibility (fibre)            | Schur floor boundary                      | cone face               |

## 6 Why this is true geometry

A Whitney-stratified orbit space is geometry in the full modern sense. It has local models at every stratum (the Luna slice), a controlled singularity structure (conical links at the walls), and its own cohomological invariants (intersection cohomology, designed for non-manifold spaces). RG is the geometry of such a bundle: a smooth Fisher-Riemannian top stratum and base, stratified spectral fibres, conical singularities at the degeneracies, and a cone boundary at the admissibility floor.

This is fully consistent with the elimination verdict. Every ingredient is established—closed Weyl chambers, Grassmannians and their Schubert strata, the PSD cone, Luna slices, monodromy and character varieties—so no new primitive is introduced, exactly as the elimination proved. What is assembled from them is a specific stratified bundle, and its coherence (the bridges tying Fisher to Schur, the second fundamental form to the transport connection, canonical correlations to the strata) is the content. The reason the object looked like it had no home is that its home is one category above manifolds, and that category is standard.

*Remark 6.1* (Answering the original intuition). The instinct that RG should not be a smooth triangle, that it should “fall back” to eigenvalues rather than paths in the blind sector, and that no manifold seemed to fit, was correct on all three counts. Paths belong to the resolved sector and the base; eigenvalues and angles belong to the coupling and null fibre; and the object is stratified, not smooth. The blind spot has no paths because it has no distinguishability—and that absence is a feature of the geometry, drawn here as the cone and the chamber wall.

## 7 Non-claims

*Non-claim 7.1.* No new mathematical primitive is introduced. Chambers, Grassmannians, PSD cones, Luna slices, and character varieties are all standard; RG is their stratified assembly.

*Non-claim 7.2.* The object is not a smooth manifold, and no smooth global chart is claimed. The stratification and its singular loci are essential, not removable.

*Non-claim 7.3.* No physical claim is made. The figures depict an estimation/observation geometry, not a physical space, and the diabolical phase is a holonomy of the null frame, not a physical field.

## 8 Conclusion

Drawing the object dissolves the paradox. RG reduces to established mathematics and yet lives in no manifold, because its natural home is a Whitney-stratified orbit-space bundle: one origin (a Schur elimination), three sector geometries (Riemannian, angular, conical), assembled into a fibre that is a product of eigenvalue chambers cut by the admissibility cone, bundled over a smooth base that carries the paths. Fisher distinguishability and the blind sector were never rivals; they are the horizontal and vertical directions of a single stratified object, meeting at the walls where eigenvalues collide and the  $\mathbb{Z}/2$  phase lives. Seen this way—and it can now be seen—RG is a geometry: not of a manifold, but of the stratified space that partial observability actually inhabits.

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